

Standard Operating Procedure for ALARM_5023: Warning - High CPU Utilization on Cisco Provider Edge Router (Cisco-PE-Delhi-NCR-16)

Node: Cisco-PE-Delhi-NCR-16 | **Vendor:** Cisco | **Severity:** Warning | **Alarm Code:** ALARM_5023

Description

This SOP outlines the steps to diagnose and resolve high CPU utilization on Cisco Provider Edge Router (Cisco-PE-Delhi-NCR-16) as indicated by Alarm Code ALARM_5023 (Severity: Warning).

Potential Root Causes

Cause	Probability
High CPU utilization due to CPU-intensive processes such as BGP, OSPF, or other routing protocols.	High
Insufficient memory or resources on the router.	Medium
Misconfiguration of CPU-intensive features or protocols.	Low

Diagnostic Commands

```
show process cpu | exclude System
show ip bgp summary
show ip ospf neighbor detail
show memory | include Total
show tech-support
```

Resolution Steps

- Step 1: {'Step': "Verify CPU utilization using the command 'show process cpu' and identify the processes consuming high CPU.", 'Action': "Use the 'show process cpu | exclude System' command to exclude system processes."}
- Step 2: {'Step': 'Check BGP and OSPF configurations for misconfigurations.', 'Action': "Review the BGP and OSPF configurations using the 'show run | include bgp' and 'show run | include ospf' commands."}
- Step 3: {'Step': 'Adjust CPU-intensive features or protocols to reduce CPU utilization.', 'Action': "Use the 'no bgp bestpath as-path multipath-relax' command to disable BGP multipath relaxation."}
- Step 4: {'Step': 'Verify memory and resource availability on the router.', 'Action': "Use the 'show memory | include Total' command to check memory usage."}
- Step 5: {'Step': 'Clear CPU-intensive processes or features.', 'Action': "Use the 'clear process cpu' command to clear CPU-intensive processes."}

Post-Check Verification

```
show process cpu | exclude System
show ip bgp summary
show ip ospf neighbor detail
show memory | include Total
show tech-support
```